MTH:2310 DISCRETE MATH
EXAM 1 REVIEW

- **Recommended Preparation**
  - Re-work examples from class.
  - Work/re-work suggested problems from the book.
  - Re-work quiz problems. Understand and learn from mistakes.
  - Go over this study guide and make sure you know and understand every point on this study guide.
  - Practice showing all your steps. Make solutions organized and be explicit/intentional in your writing.
  - Replicate the exam environment. Can you do relevant problems with little to no help?

1.1 **Propositional Logic**

  - Be able to define proposition.
  - Be able to determine whether a statement is a proposition and state its truth value.
  - Be able to explain why a statement is or is not a proposition.
  - Be able to convert a given compound proposition from an expression involving propositional variables and logical connectors into words, and from words into an expression involving propositional variables and logical connectors.
  - Be able to complete a truth table for a compound proposition.

  Ex. Section 1.1, Exercises 1 - 3 (additionally explain why the statements are or are not a proposition), 10 - 12, 15, 34

1.3 **Propositional Equivalences**

  - Be able to determine when two compound propositions are logically equivalent, and explain why.
  - Be able to determine when a compound proposition is a tautology, contingency, or contradiction.
  - Be able to define tautology, contingency, and contradiction.
  - Use logical equivalences to simplify or rewrite a compound proposition.

- Be able to determine whether a compound proposition is satisfiable.
  (a) If it is satisfiable, know how to find an example of an assignment of truth values which satisfies the compound proposition and show that it satisfies the statement.
  (b) If it is unsatisfiable, know how to justify why there is no assignment of truth values which satisfies the compound proposition.

Ex. Section 1.3, Exercises 9 - 12, 20, 21, 33, 34 65, 66

1.4 **Predicates and Quantifiers**

- Be able to identify predicates and subjects within a statement.
- Be able to convert a sentence into a compound proposition using propositional functions, quantifiers, and logical connectors.
- Be able to convert a compound proposition given as an expression using propositional functions, quantifiers, and other logical connectors into a sentence (or sentences).
- Be able to determine the truth value of a quantified propositional function based on the indicated domain.
- Be able to determine logical equivalences between quantified statements.

Ex. Section 1.4, Exercises 5, 6, 9, 10, 13, 14, 23, 24